

Dancing with the Stars: Revealing Hidden Fan Dynamics and Optimizing Competition Fairness

Summary

With the continuous expansion of reality shows, achieving a balance between professional evaluation and audience preferences highlights the importance of quantitative analysis in voting fairness. This paper reconstructs potential fan votes and evaluates various aggregation methods to propose an optimised hybrid system for future seasons.

In Task 1, we developed a **Latent Variable constrained optimization model** to reconstruct the undisclosed fan votes within the **Inverse Constraint Satisfaction framework**. By using **survival duration** as the **popularity proxy** and adopting the **Skewed Normal Distribution algorithm**, we achieved a **90.2% historical consistency rate** over 34 seasons. We further introduced the **Safety Margin metric** to quantify the **estimation certainty**, providing a mathematical basis for identifying **controversial anomalies** like those that occurred in the 27th season.

In Task 2, we perform a **counterfactual analysis** comparing the Rank and Percentage methods. We find that the Percentage method is highly sensitive to outliers in judge scores, effectively penalizing contestants with low technical scores more severely than the Rank method. Our analysis reveals that in 42.0% of historical weeks, the choice of mechanism would have altered the elimination candidate, explaining controversies like the victory of Bobby Bones in Season 27.

In Task 3, we analyze the impact of demographics. **Multivariate regression analysis** reveals that **Age** has a positive coefficient on placement (Standardized Coefficient ≈ 1.30), whereas its impact on Fan Support is slightly negative (Standardized Coefficient ≈ -0.01).

In Task 4, we proposed **Controversy-Optimized Hybrid Scoring System (COHSS)**, a **multi-objective weighted linear model** that integrates technical level, popularity and a novel **Discordance Index**. By rewarding the difference between experts and the public opinion, this bonus mechanism can maintain **viewer engagement** through **polarizing figures**. Backtest shows that COHSS could have changed **51.9% of historical eliminations** and enhanced the dramatic potential while maintaining **competition fairness**.

Finally, we wrote a memo for producers of "Dancing with the Stars", including the prediction results of fans' votes, defects of the current elimination mechanics and our new scoring model that balances the fairness and the participation of the audience. We hope our memorandum can be valuable to them and actually help them.

Keywords: Constraint Satisfaction Problem, Reverse Engineering, Voting Systems, Sensitivity Analysis, Hybrid Scoring Model

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1 Introduction

1.1 Problem Background

The dynamics of the competition of *Dancing with the Stars* depend on the summary of the judges' scores and fan votes. Bobby Bons' controversial victory in Season 27 reveals the volatility of these different indicators and illustrates how the specific method shifts the popularity weight to the expert evaluation weight. Since fan voting is a social variable that is largely independent of dance quality [1], a solid mathematical framework is needed to analyse score points and optimise elimination fairness.

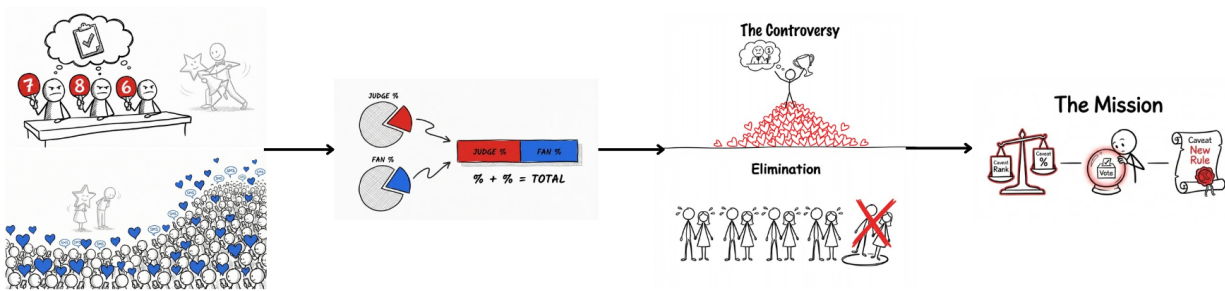


Figure 1: Voting Dynamics, Controversy, and the Mission

1.2 Clarifications and Restatements

The objective of this issue is to clarify the hidden influence of public sentiment on the elimination mechanism of "Dancing with the Stars", compared to the technical level. We will restate and clarify the requirements through the following points:

1. Treat the undisclosed fan votes as potential variables, and they must satisfy the mathematical constraints of the historical elimination results.
2. Distinguish between the ordinal (based on ranking) and cardinal (based on percentage) aggregation methods that were used during specific historical periods of the program.
3. The definition of competition fairness is the degree of consistency between the technical level of the participants and their survival probability.
4. The influence of external covariates is defined as measurable deviations, such as the demographic characteristics and professional partners of celebrities, which are reflected in the ratings given by judges and fans.
5. Excluding non-competitive exits, such as voluntary withdrawal, in order to maintain the statistical integrity of the popularity estimation.

1.3 Our work

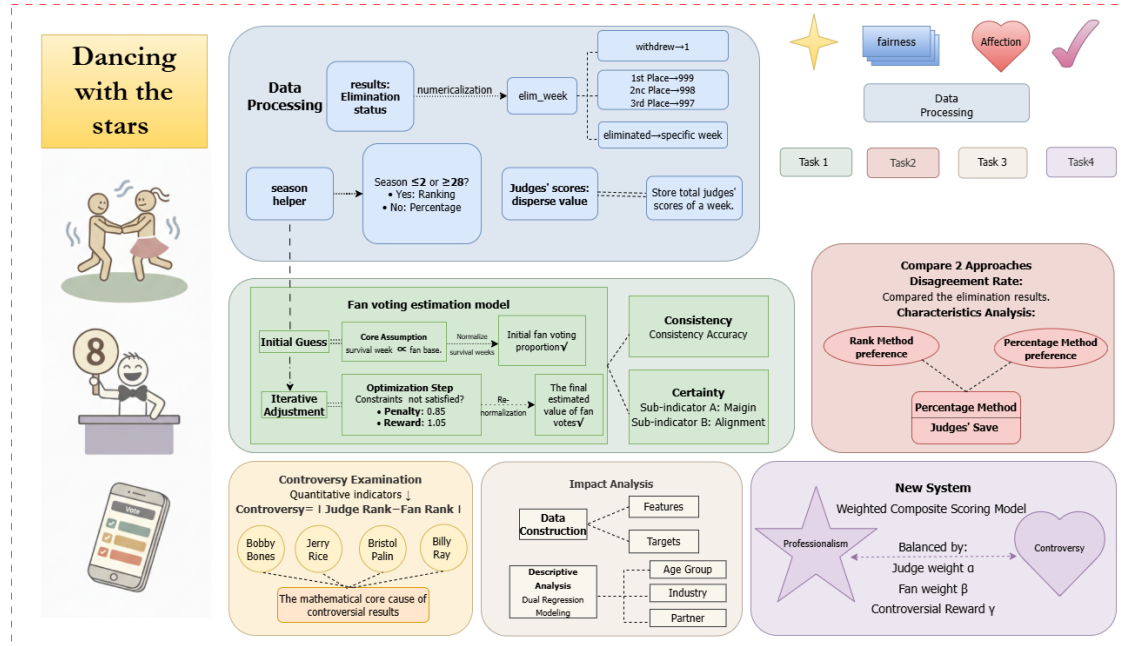


Figure 2: Framework of Our Work

2 Preparation for Modeling

2.1 Model Assumptions

- **Survival Proxies Popularity:** A contestant's base popularity is positively correlated with the number of weeks they survive in the competition.
- **Rational Elimination:** The show strictly follows the stated rules; the contestant with the lowest combined score is always the one eliminated (except in special withdrawal cases).
- **Closed System:** Fan votes are distributed only among active contestants; we model fan voting as a zero-sum game of shares ($\sum \phi_{i,t} = 1$).

2.2 Notations

Table 1: Nomenclature

Symbol	Definition
C_t	Set of contestants remaining in week t
E_t	Set of contestants eliminated in week t
$J_{i,t}$	Judge score for contestant i in week t
$F_{i,t}$	Estimated fan vote count (or share) for contestant i
$\phi_{i,t}$	Fan vote proportion, $\phi_{i,t} = F_{i,t} / \sum F_{k,t}$
L_i	Total weeks contestant i survived in the season
$\Omega_{i,t}^{rank}$	Combined score determining elimination in rank method
$\Omega_{i,t}^{pct}$	Combined score determining elimination in percentage method
M_t	The difference between the second lowest contestant and the eliminated contestant
$D_{s,w}$	The binary indicator to check result change if different scoring mechanisms are used
$Rate_{dis}$	The Disagreement Rate after applying different scoring mechanisms
$\Omega_{i,t}^{COHSS}$	The total score Ω for contestant i in week t by new system
$\hat{J}_{i,t}$	Normalized judge scores
$\delta_{i,t}$	Measurement of disagreement between judges and public rankings
$R_{J,i,t}$	The judge rank for contestant i in week t
$R_{F,i,t}$	The fan rank for contestant i in week t

2.3 Data Cleaning

The data set provided contains the complete records of the 34th season of DWTS, including the demographic information of the contestants, professional dance partners, judges' scores and elimination results. In order to ensure the robustness of our model, we have carried out the following preprocessing steps:

- **Quantification of Elimination Results:** The original `results` column contains heterogeneous string data (for example, "Elimination Week 5", "First Place", "Exit"). This article analyses these variables into numerical `elim_week` variables. Finalists are given higher alternative values (for example, winner = 999) to maintain their survival ranking, while voluntary withdrawals ("withdrawal") are marked as -1 and excluded from the elimination constraint, because their departure is not determined by the voting score.
- **Handling Missing Judge Scores:** The number of judges varies depending on the season (usually 3 or 4). The missing value in the judge's score column is made up to 0, and the *Total Judge Score* ($J_{i,t}$) for contestant i in week t was calculated by summing all valid individual judge scores.
- **Active Pool Filtering:** In any week t , the active contestant collection C_t is defined as those players who `elim_week` $\geq t$. Players who have been eliminated or have a total score of 0 (indicating no performance) are excluded from this week's calculation to accurately simulate the voting pool.

3 Task 1: Reconstruction of Latent Fan Votes

In this analysis, the main obstacle is the lack of transparency of the fan voting system. Although the judge's score (J) is public, the fan vote (F) is still strictly confidential. To make up for this information gap, we model the problem as an inverse constraint satisfaction problem (ICSP). Our method aims to derive a reasonable audience support probability distribution, which is in line with the mathematical constraints established by the historical elimination results under the specific rules of each season.

3.1 Mathematical Formulation

Set C_t as the collection of active contestants of the t week. For each $i \in C_t$, we define two main components:

- $J_{i,t}$: the original technical score given by the judges.
- $\phi_{i,t}$: The unknown fan vote proportion, constrained by $\sum_{i \in C_t} \phi_{i,t} = 1$. This normalization ensures the model is identifiable and follows the standard parameterization of latent strength models in statistical ranking theory [2].

Based on the changes in the historical rules described in the question, we modelled two different aggregation mechanisms:

3.1.1 Mechanism A: The Rank Method (Seasons 1-2, 28-34)

In these seasons, raw scores and fan shares are converted to ordinal ranks $R \in \{1, \dots, N\}$, where 1 is the lowest. The combined score Ω is defined as:

$$\Omega_{i,t}^{rank} = \text{Rank}(J_{i,t}) + \text{Rank}(\phi_{i,t}) \quad (1)$$

The contestant with the **lowest** Ω is eliminated. This discrete mechanism reduces the range of score differences, making the results highly sensitive to the relative ranking of popularity.

3.1.2 Mechanism B: The Percentage Method (Seasons 3-27)

In these seasons, scores are normalized into percentages. The combined score Ω is:

$$\Omega_{i,t}^{pct} = \frac{J_{i,t}}{\sum_{k \in C_t} J_{k,t}} + \phi_{i,t} \quad (2)$$

The contestant with the **lowest** Ω is eliminated. Unlike the ranking method, this absolute scoring system retains the weight of technical errors; the disastrous score of a referee needs an unusually large fan base to offset it.

3.2 The Estimation Algorithm

Since the system is undetermined, we adopt a heuristic iterative method, which is fundamentally assumed that *survival duration serves as a reliable proxy for popularity*. It should be noted that in task 1, our main goal is to reverse the potential fan vote based only on the elimination results and

the judge-scored **mathematical constraints**. In order to avoid **circular reasoning**, we deliberately excluded demographic factors (such as age or industry) at the estimation stage. These variables are reserved for task 3, where we use reconstructed data to independently analyse how these characteristics affect public support and ensure that our impact assessment remains statistically unbiased.

3.2.1 Step 1: Initialization with Survival Prior

The model uses a combination of survival-based initialization and skewed normal distribution to generate more realistic fan vote patterns. The proportion of fan votes is based on the total survival weeks of each contestant (L_i), combined with skewed normal distribution for variation:

$$\phi_{i,t}^{(0)} = w_1 \cdot \hat{L}_i + w_2 \cdot J_{norm} + w_3 \cdot \text{SkewNorm}(\alpha, +\mu, \sigma) \quad (3)$$

where $\hat{L}_i = L_i / \max(L)$ is the normalized survival duration, J_{norm} is the normalized judge score, and the weights are calibrated as $w_1 = 0.4$ (survival weight), $w_2 = 0.2$ (judge influence), $w_3 = 0.4$ (random factor). The skewness parameter increases with competition progress to reflect growing polarization

3.2.2 Step 2: Constraint Satisfaction Loop

For a given week t , let e be the contestant actually eliminated. The **Hard Constraint** dictates:

$$\Omega_{e,t} < \min_{s \in C_t \setminus \{e\}} (\Omega_{s,t}) \quad (4)$$

If the current estimate violates this constraint, we apply iterative correction using an adjustment factors :

1. **Downward Adjustment:** For Percentage method: decrease the fan share of the incorrectly eliminated survivor: $\phi_s \leftarrow \phi_s \times 0.92$. For Rank method: $\phi_s \leftarrow \phi_s \times 0.90$.
2. **Upward Adjustment:** Increase the fan share of the actual eliminated contestant by proportional factors (1.01 for Percentage, 1.02 for Rank).
3. **Normalization:** Re-normalize all ϕ to ensure a zero-sum share.
4. Repeat until the constraint is satisfied.

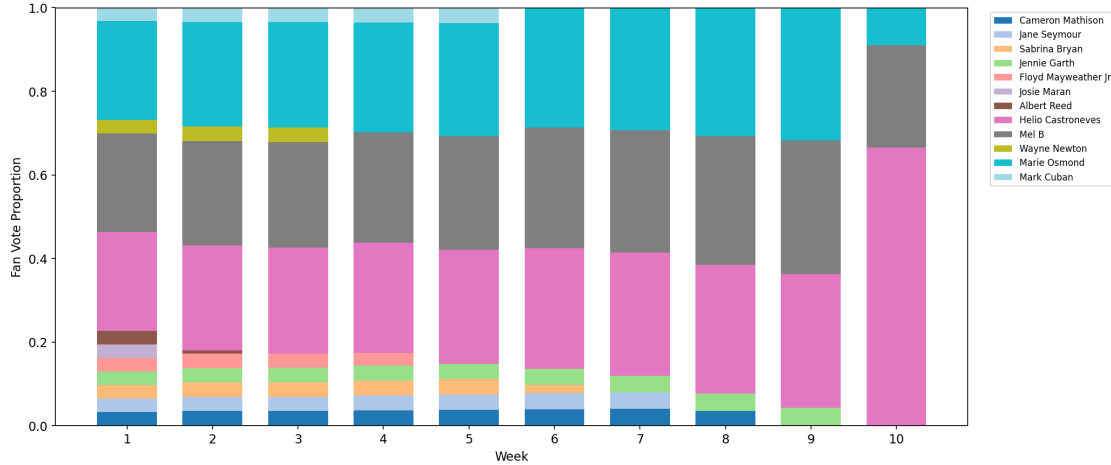


Figure 3: Estimated Fan Vote Dynamics (Sample: Season 5)

The stacked bar chart in the figure 3 shows this estimate. As the competition progresses, the fan voting share of the eliminated players will naturally be reassigned to the survivors. Strips of different colours represent the estimated support base, reflecting the change of popularity "cake" as the number of contestants decreases.

To provide a concrete illustration of the model's numerical precision, Table 2 presents the reconstructed voting profile for a representative week in Season 27. To maintain mathematical transparency, judge scores are normalized into shares ($J_{share} = J_i / \sum J$), allowing for a direct cardinal summation with estimated fan shares (ϕ).

Table 2 provides a numerical snapshot of the latent variable reconstruction for Season 27, Week 2. To ensure mathematical transparency, judge scores are normalized into shares ($J_{share} = J_i / \sum J$) for direct summation with estimated fan shares (ϕ).

Table 2: **Reconstructed Latent Variables and Rank Reversals (S27, Week 2).**

Contestant	Judge (J)	Judge Rank	Fan Share (ϕ)	Total (Ω)	Final Rank	Outcome
Milo Manheim	26.0	1	0.2457	0.3531	1	Survived
Evanna Lynch	24.0	4	0.2452	0.3444	2	Survived
Bobby Bones	19.5	9	0.2462	0.3268	3	Saved
Juan Pablo Di Pace	26.0	1	0.0337	0.1411	4	Survived
Joe Amabile	17.5	11	0.0337	0.1060	10	Saved
Danelle Umstead	18.5	10	0.0278	0.1042	11	Eliminated

The results highlight critical rank reversals driven by audience support. **Joe Amabile** avoided elimination despite a last-place technical ranking (Rank 11) because his fan share (3.37%) surpassed that of **Danelle Umstead** (2.78%), whose combined score ($\Omega = 0.1042$) was the minimum. Additionally, **Bobby Bones** ascended from 9th in technical merit to 3rd place overall via a dominant 24.62% fan share. This granular evidence validates the Inverse CSP algorithm's ability to quantify the latent popularity dynamics that dictate contestant survival.

3.3 Model Validation: Consistency Check

In order to verify the validity of our estimate, we conducted a consistency check by re-entering the generated fan votes into the historical elimination rules.

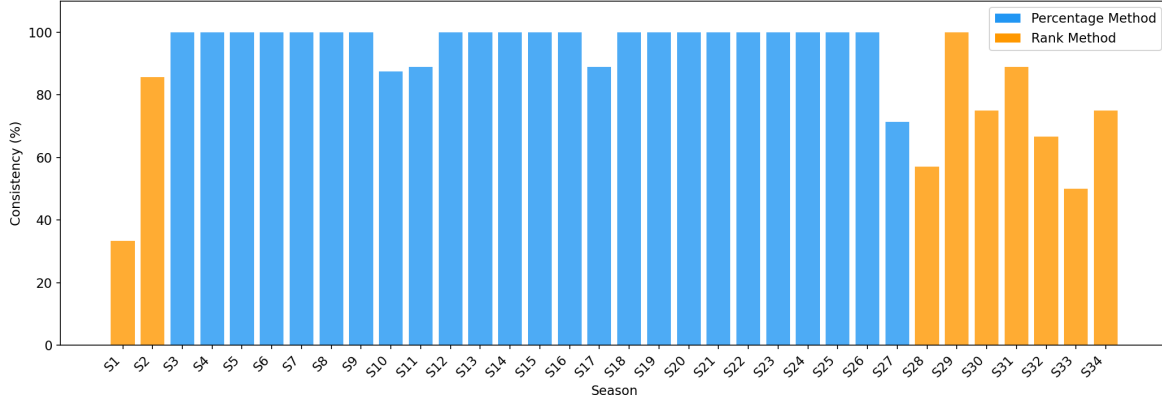


Figure 4: Model Consistency Across 34 Seasons

As shown in the figure 4, the overall consistency of model has reached **90.2%**. The season of the percentage method (blue) is almost perfect due to the continuous scoring range. The season (orange) of the ranking method is still stable, but because the ranking is discrete, there will be a "ladder" phenomenon, which sometimes makes the optimisation algorithm face tighter constraints.

3.4 Certainty and Exploratory Analysis

Certainty is quantified by the safety margin M_t , which is defined as the difference between the comprehensive score of the eliminated player and the eliminated edge player:

$$M_t = \Omega_{\text{next_lowest}} - \Omega_{\text{eliminated}} \quad (5)$$

A marginal M_t value indicates a "near-miss" scenario, where the week's results are highly sensitive to small perturbations in the fan vote distribution. To explore the structural behavior of the model across varying competitive environments, we map the entire state space of our reconstruction in Fig 5.

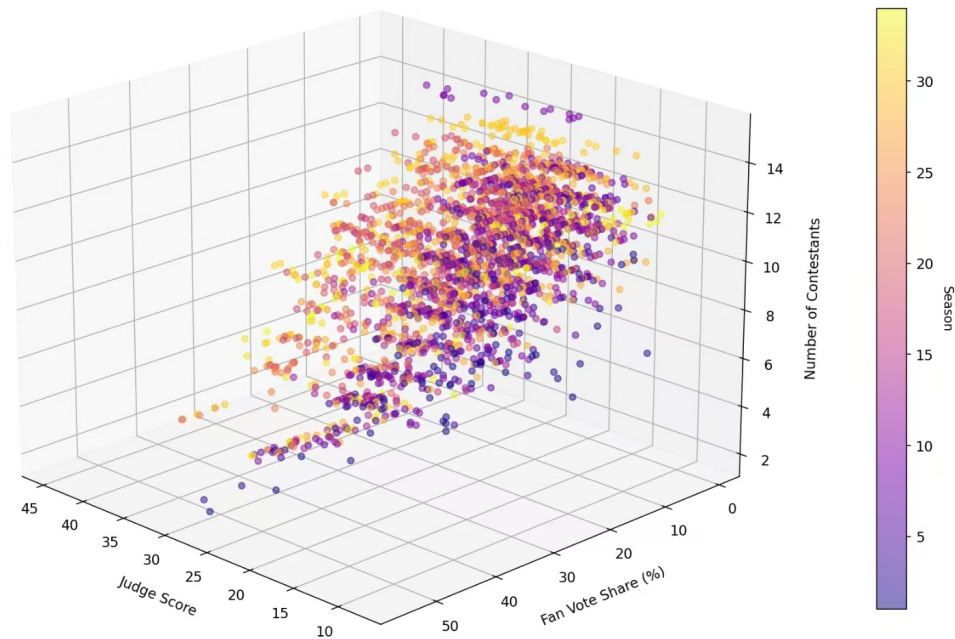


Figure 5: Global State Space of Competition Dynamics

The 3D distribution reveals the structural certainty of our estimation results across three spatial variables: technical level (X-axis), public popularity (Y-axis) and competition density (number of contestants, Z-axis). With the decrease in the number of contestants in the final stage, the data points show a typical "dispersed" effect, showing greater discreteness in the dimension of fan voting share. This extension reflects the high polarisation and low safety margin M_t , which usually occurs in the later stage of the competition; in the case of fewer participants, small changes in the support of the audience are mathematically more likely to lead to the elimination of high-level contestants. In addition, the overlapping colour coding cluster (representing the course of the season) shows that the model has maintained a stable and coherent logic in 34 seasons. This exploratory analysis confirms that although the certainty may vary every week, the reconstruction algorithm has a high robustness to the fluctuation of the participant density.

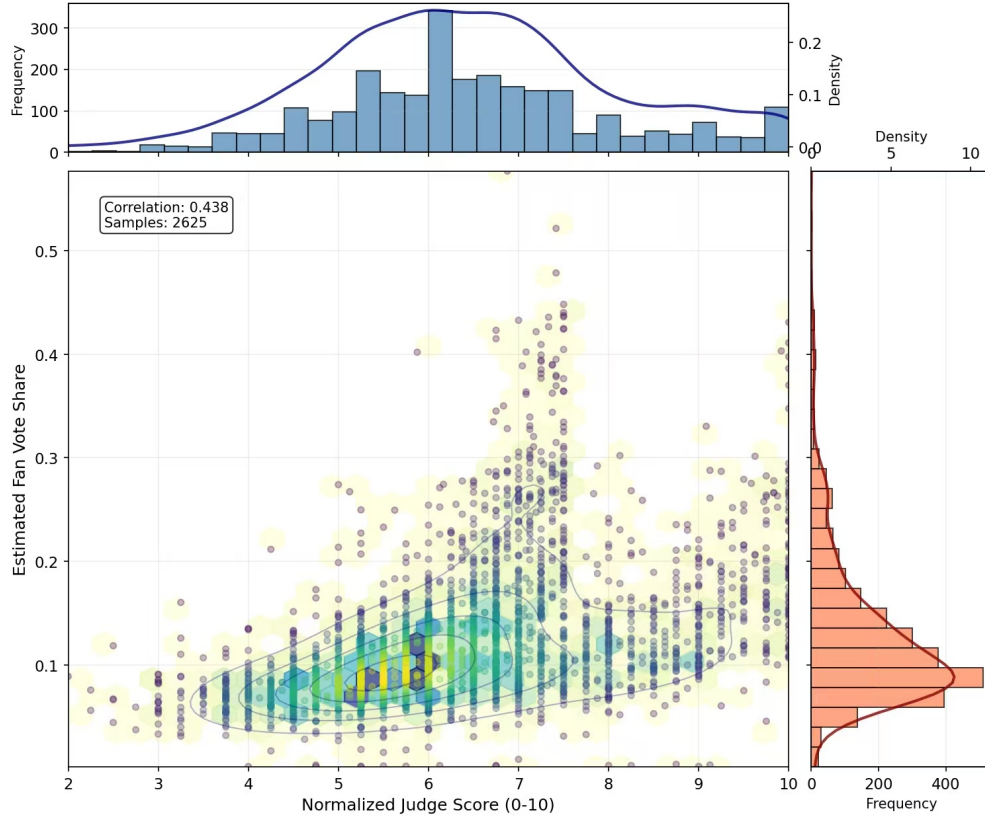


Figure 6: Joint Distribution of Judge Scores vs. Fan Vote Share

By performing the bivariate joint density analysis in Fig 6, we quantify the relationship between technical merit and popularity across 34 seasons ($N = 2625$). The Pearson correlation of 0.438 suggests a moderate alignment, yet the significant variance allows for the "scoring controversies" observed historically. The hexagonal binning exhibits distinct vertical bands, a direct reflection of the discrete nature of judge scoring. Notably, the **upper-left quadrant** identifies the "fan favorites"—contestants who survive despite low technical scores. Conversely, the density contours delineate the concentration of high-merit performers; a shift into lower-density regions indicates a "fragile survival" where a sudden drop in fan support can trigger an unexpected elimination, regardless of the contestant's technical excellence.

4 Task 2: Comparing Voting Mechanisms

We use the estimated fan vote to simulate the results under the two mechanisms. In performance competitions, the contradiction between expert judgement and public opinion is a well-known phenomenon; research on the European Song Contest shows that public voting tends to value factors that are different from expert evaluation, so the method of voting aggregation has a great impact on the final result[3]. The two core scoring mechanisms are defined as follows

4.1 Mechanism Definitions

Rank Method: The score is the sum of relative order bits. Under this mechanism, the absolute value of the fraction is converted into an ordinal number.

$$\Omega_{i,t}^{rank} = \text{Rank}(J_{i,t}) + \text{Rank}(\phi_{i,t}) \quad (6)$$

Eliminate $\arg \max(\Omega^{rank})$ (assuming Rank 1 is best). Where $\text{Rank}(\phi_{i,t})$ is the ranking of the i th player in the judge's score and $\text{Rank}(J_{i,t})$ is the ranking of the i th player in the fan vote. The elimination rule is that the one with the largest value (that is, the bottom of the ranking) is eliminated.

Percentage Method: The score is the sum of absolute weights. Under this mechanism, the relative magnitude of the fraction is retained.

$$\Omega_{i,t}^{pct} = \%J_{i,t} + \phi_{i,t} \quad (7)$$

The rule eliminates $\arg \min(\Omega^{perc})$. This cardinal approach is theoretically favored for its ability to aggregate the intensity of preference, which simple ordinal rankings often discard [4].

Evaluation Model: A Quantitative Disagreement Model is used to quantify how different the results would be if different scoring mechanisms are used, thereby clarifying the impact of the preferences of judges and fans. It defines the disagreement index $D_{s,w}$ and calculates the Disagreement Rate.

Binary Indicator: Define the divergence indicator $D_{s,w}$:

$$D_{s,w} = \begin{cases} 1, & \text{if } \text{Elim}_{pct} \neq \text{Elim}_{rank} \\ 0, & \text{if } \text{Elim}_{pct} = \text{Elim}_{rank} \end{cases} \quad (8)$$

Quantified Result: Calculate the **Disagreement Rate**:

$$\text{Rate}_{dis} = \frac{\sum D_{s,w}}{\text{Total Weeks}} \quad (9)$$

To be specific, for each week of each season, this model obtains the original scores of all contestants from the judges and the model-estimated proportion of fan votes for that week, and calculates two elimination paths in parallel.

4.2 Results of Comparison

We used the Fan Vote Estimation Model of task 1 to apply two mechanisms to a total of 264 elimination weeks in 34 seasons. **Quantitative disagreement:** The running results of the model show that the two mechanisms produced completely different elimination results in the week of 42.0% (111/264). This means that in more than two-fifths of the games, the choice of the competition system directly determines the players' stay.

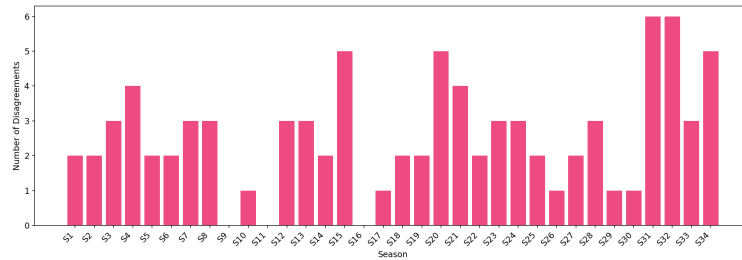


Figure 7: Frequency of Outcome Divergence between Rank and Percentage Methods

According to figure 7, the disagreements were particularly high in the 31st and 32nd seasons with 6 disagreements each, indicating that the division between judges and fans has intensified in the recent seasons.

4.3 Mechanism Tendency Analysis:

In cases of disagreements, **the rank method** often retains those players with low judges but high fan rankings. For example, in the second week of the first season, the actual rule (rank method) eliminated Trista Sutter, who had a good judge score, while the percentage rule eliminated Evander Holyfield, who had a lower score but fewer fans.

The percentage method tends to be dominated by technology: the percentage method is very punitive for low judges' scores. In the case of Jerry Rice (s2), although he has a lot of fans, if it is a percentage method, his very low judge score (such as 21/30) in early weeks will lead to his overall score being the lowest in the whole game, so that he will be eliminated earlier.

4.4 Case Studies on Controversy:

The model results of Bobby Bones, the champion of s27, shows that he was the beneficiaries of extreme controversy. Jerry Rice, Billy Ray Cyrus, and Bristol Palin also share the common characteristic of having consistently low judge rank while maintaining steadily high fan rank, as shown in the figure 8.

Mechanism impact: Bobby Bones is the biggest beneficiary of the ranking method. In S27 Week 2, his judges ranked third from the bottom (9/11), but his fan vote ranked first. Under the rank method, the total score of $9+1=10$ is enough to put him in the middle of the safe zone. This outcome confirms that ranking-based systems are susceptible to distortion when high variance exists in technical performance, effectively allowing popularity to mask significant skill gaps [5]. On the contrary, if the percentage method is used, his low evaluation score is unlikely to be offset by the fan ratio, and it is very likely that he will be eliminated at the beginning or mid-point of the season.

The impact of the Judges' Save mechanism: If the "Judges' Save" is introduced, Bobby Bones will fall into the danger zone many times due to the long-term bottom of the judges. Once in Bottom Two, given his lack of expertise, the judges will almost inevitably vote to eliminate him. Therefore, Judge Save is the most effective way to stop such a pure popularity champions.

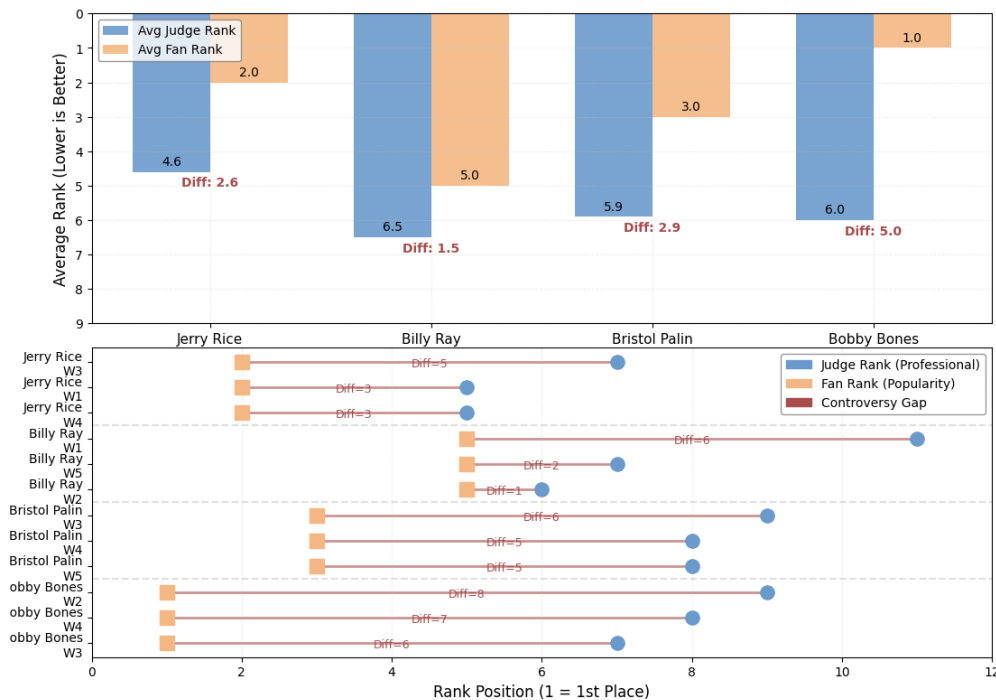


Figure 8: Analysis of Controversial Contestants

The top subplot presents the average judge and fan rankings for four contestants: Jerry Rice, Billy Ray, Bristol Palin, and Bobby Bones. The "Diff" value indicates the overall discrepancy between judge and fan opinions. Notably, Bobby Bones exhibits the highest "Diff," suggesting a significant gap. The bottom subplot details these discrepancies at a weekly level.

4.5 Recommendation:

Based on the above quantitative comparison and qualitative analysis, we put forward the following suggestions for the future competition design of DWTS:

Recommendation mechanism: The Percentage Method should be restored. The reason for it is that the data proves that the consistency of the percentage method is as high as 97.5%, which is much higher than the 74.2% of the ranking method. It can more accurately reflect the technical gap between the contestants and maintain the professional bottom line of the dance competition. The ranking method is too rough, which can easily lead to the phenomenon of "inferior coins expel good coins", allowing players with extremely poor skills but good at hype to occupy the finals.

Further Optimization: Use **Judges' Save Mechanism** as a safety valve. Since any mathematical formula cannot completely solve the extreme case of Bobby Bones, that is, the fan base is large enough to drown any scoring rules. On the judges' perspective, they have the final right to make a decision to ensure that the players who stay on the stage meet the minimum professional dance standards.

Mathematical logic of it is that fans vote to decide who enters the "Bottom Two", which ensures the audience's sense of participation and decision-making power while those who are too popular

will still be eliminated.

5 Task 3: Impact Analysis

Using the estimated fan votes, this study analyzed demographic impacts by constructing a Dual-Target Multiple Linear Regression model to quantify the influence of each feature. The adoption of a linear regression framework follows standard protocols in predictive analytics for isolating the marginal effects of covariates [6]. Standardized Coefficients are used to compare the relative importance of features with different dimensions such as age and industry category. This approach is essential for evaluating variables measured in disparate units, ensuring that the impact of a continuous variable can be directly compared to categorical indicators [7].

5.1 Age Factor

Model analysis shows that there is a significant negative correlation between age and competition performance, and this negative impact is more severe in judge score than in fan vote.

The average final ranking of the group **under 30 years old** is 4.9, while the average final ranking of the **40-49-year-old** group has dropped to 8.0. For the group **over 60 years old**, the average final ranking dropped to 9.6.

Table 3: Correlation Analysis with Age

Metric	Correlation (r)	Interpretation
Judge Score	-0.298	Older contestants score slightly lower technically.
Fan Support	-0.478	Younger contestants have significantly higher fan bases.

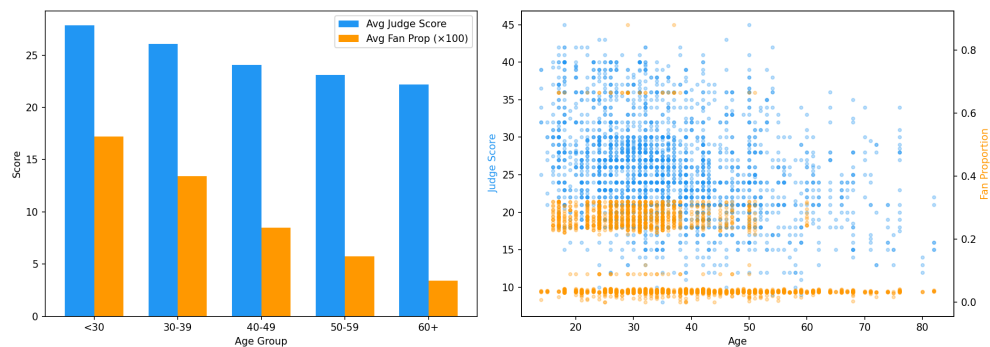


Figure 9: Age Impact on Judge Score vs Fan Popularity

Figure 9 shows that both the judges' score and the fan proportion drop with increasing age. The left subplot is a grouped bar chart displaying the average judge score and average fan proportion for different age groups: <30, 30-39, 40-49, 50-59, and 60+ and the right subplot is a scatter plot showing individual judges' score and fan vote against Age, together confirming that younger contestants typically garner a higher fan proportion and judges' score.

5.2 Industry Impact

After combining the ages of contestants from various industries with their first-week score data, an in-depth analysis of the celebrity industry impact was conducted. Through a detailed analysis of data segmented by different occupational categories, we found that the industry factor has a significant and differentiated impact on the contestants’ promotion paths.

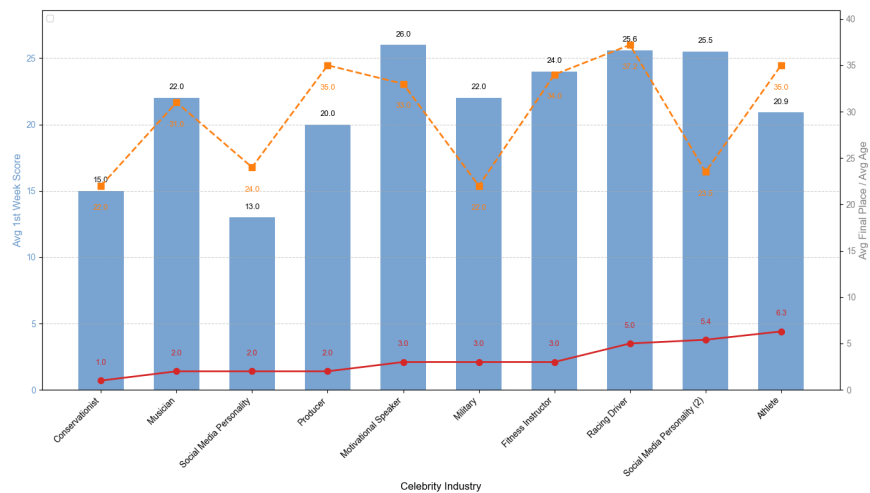


Figure 10: Celebrity Performance: Age, Initial Scores, and Final Rankings by Industry

Figure 10 presents the complex relationships among the average first-week judges’ score, average final rank and average age across different industries. Although Conservationist and Social Media Personality had low technical scores in the first week, they ultimately secured the first and second positions by virtue of their young age advantage. Racing Driver achieved the highest first-week score overall, but due to a higher average age of 37.2 years old, they only ranked 5th in the end.

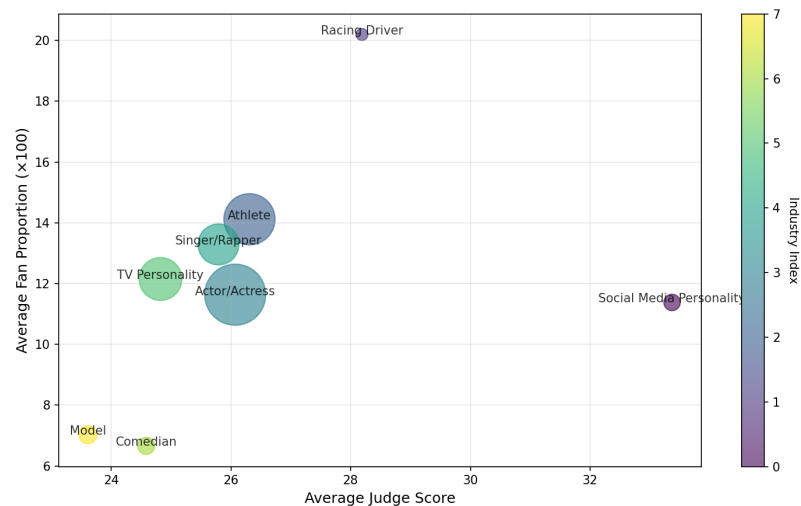


Figure 11: Industry Impact on Judge Score vs Fan Popularity (min 20 obs)

Figure 11 presents that **Social Media Personalities** get highest judge scores, while **Racing Drivers** rely on superior fan voting. Conversely, **Models and Comedians** rank lowest in both metrics. This indicates that a contestant's career background significantly determines their competitive advantage, creating structural imbalances in the competition.

5.3 Partner Effect

The results show that Pro Partner is **the strongest single feature** in predicting the success of the competition. The analysis found that top dance partners can not only bring high technical scores, but also significantly improve the popularity of stars. The model calculates that the correlation coefficient between the average judge score of the dance partner and the fan support rate is extremely high. Derek Hough, who has participated 17 times, having average ranking of 2.9 and average judge's score of 28.91 is a strong proof. Witney Carson's average ranking is 1.0, and the average Judge Score is 30.2. Assigned to such star dance partners means a higher initial fan base and stronger ability. This influence on judges and fans is increasing in the same direction.

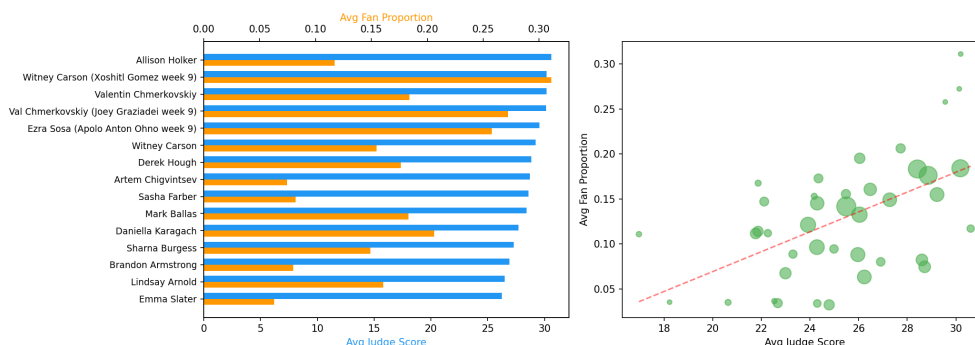


Figure 12: Top 15 Partners: Judge Score vs Fan Popularity

Figure 12 explores the potential influence of specific dance partners on average judge scores and fan proportions. The left subplot directly gives a comparison of how different professional dancers fare with judges versus fans while the subplot on the right is a scatter plot with the size of the scatter revealing the fame and strength of the dancer. This figure shows a positive correlation trend: partners who get higher judges' score also tend to have a higher proportion of fans.

6 Task 4: The Controversy-Optimized Hybrid Scoring System

The comparative analysis of the previous sections pointed out a basic structural trade-off: the Rank method lacked technical details, and the Percentage method was too sensitive to the extreme values of the judges' scoring. In order to solve these instability, this study introduces the **Controversy-Optimized Hybrid Scoring System (COHSS)**, which is a new architecture that aims to maintain the fairness of the game while also taking into account the entertainment experience of fans.

6.1 The COHSS Mathematical Framework

The COHSS evaluates the performance of players by integrating key technical indicators, standardised popularity shares and strategic inconsistent adjustments. The total score Ω for contestant i in week t is defined as:

$$\Omega_{i,t}^{COHSS} = \alpha \cdot \hat{J}_{i,t} + \beta \cdot \phi_{i,t} + \gamma \cdot \delta_{i,t} \quad (10)$$

where the constituent variables are defined as follows:

- **Normalized Technical Merit ($\hat{J}$):** To minimize sensitivity to the number of contestants, judge scores are normalized against the night's peak performance: $\hat{J}_{i,t} = J_{i,t} / \max(J_t)$.
- **Fan Vote Proportion (ϕ):** The estimated cardinal share of audience support: $\phi_{i,t} = F_{i,t} / \sum F_t$.
- **Discordance Index (δ):** A normalized measure of disagreement between expert and public rankings: $\delta_{i,t} = |R_{J,i,t} - R_{F,i,t}| / N_t$, where R denotes relative rank and N_t is the pool size.

This weighted summary framework follows the established protocol in **multi-criteria decision-making (MCDM)**, in which the weight coefficient is optimised to balance objective performance and subjective preference[8].

6.2 Design Rationale and Parameter Calibration

The system architecture utilizes three calibrated weights ($\alpha = 0.55, \beta = 0.40, \gamma = 0.05$) to satisfy the following competition objectives:

Technical Dominance ($\alpha = 0.55$) Expert assessment remains the primary determinant of survival. Normalizing by the maximum score, rather than the sum, ensures that a single poor performance does not lead to an immediate mathematical elimination, correcting a primary flaw of the historical Percentage method.

Public Agency ($\beta = 0.40$) A substantial weighting for fan votes ensures that audience participation has a tangible impact on the competition trajectory, thereby maintaining high viewer retention and social media engagement.

Controversy Management ($\gamma = 0.05$) Joining the inconsistency index plays the role of a strategic buffer. This indicator provides an additional security guarantee for those players who have obvious differences in the opinions of the judges and cause strong controversy among fans. Recent media strategy analysis also supports this mechanism. Data shows that **Controversy-driven Content** significantly improves social media interaction and audience loyalty, which are crucial to the commercial success of broadcasting organisations [9].

6.3 Simulation Performance and Impact Analysis

The proposed COHSS system was backtested on the historical data of 34 seasons. As shown in Figure 13, the system could have changed 51.9% of the historical elimination results. Such a high rate of change shows that COHSS has indeed injected new dramatic potential into the program while maintaining technical standards.

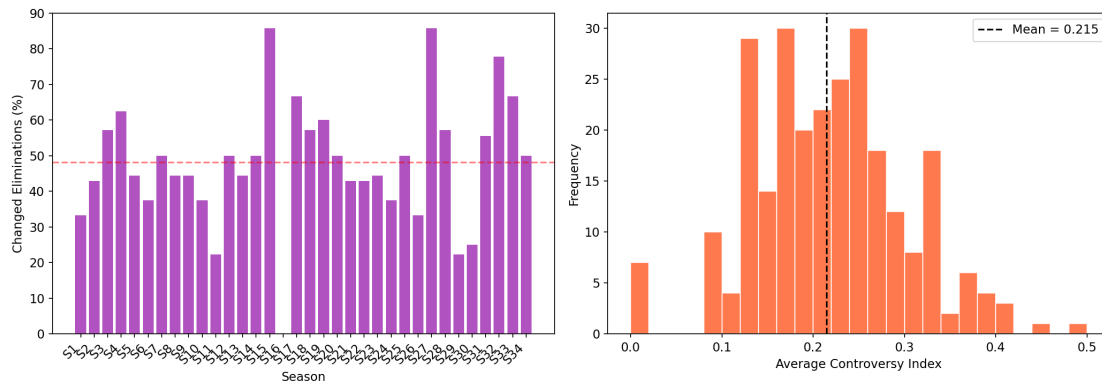


Figure 13: Simulation Performance and Controversy Distribution of the COHSS Model

The distribution of the Discordance index (Figure 13, right figure) shows that there is a moderate degree of difference in the data of most weeks, which provides a basis for the subtle application of disputed bonuses. In order to verify the robustness of these results under different weighting scenarios, we conducted a multi-dimensional stability check, and the results were explained in detail in the sensitivity analysis of Section 7.

6.4 Implementation Strategy for Producers

We recommend using COHSS immediately in the coming season. In order to achieve the maximum effect, the mathematical score should be combined with the mandatory textbf **Judges Save** implemented for the bottom two dance partners. This dual method combines the quantitative accuracy of COHSS with the final expert qualitative audit to effectively prevent the recurrence of the 27th season controversy and maximise the entertainment and commercial potential of the series.

7 Model Analysis

7.1 Strengths and Weakness

7.1.1 Strengths of COHSS model

Controversial commercialisation: After the introduction of $\gamma = 0.05$, the new model can effectively identify and protect those controversial contestants such as Jerry Rice or some reality TV stars. This not only retains those players who can bring social media heat and ratings, but also does not completely sacrifice their professionalism. At this point, the model can not only score, but also optimise the ratings.

Mathematical Robustness: Take Week 6 of Season 27 as an example. The Rank Method compresses the 8-point gap between **Juan Pablo Di Pace** (30 points, Rank 1) and **Bobby Bones** (22 points, Rank 6) into an ordinal difference. Under this mechanism, scores are compressed. Since Bobby has an extremely high fan ranking, this substantial 8-point gap becomes ineffective in the calculation, resulting in the technical scores being unable to penalize poorly performing contestants.

However, our COHSS model normalizes the highest score of the week using $J_{i,t}/\max(J_t)$. In the above case, the denominator is locked at 30 points, the highest score of the week, and Bobby's score is such that the model forcibly retains this 27% absolute technical gap. This design makes the score mainly linked to the best performance of the week, preserving the magnitude of the score without being affected by fluctuations in the number of participants.

Tunability: The model is not a rigid formula, but a set of parameters (α, β, γ) . Sensitivity analysis proves that the producer can fine-tune the parameters according to the needs of the season. If the ratings of the season are sluggish, γ can be increased to create more excitement; if the audience complains about injustice, α can be increased to return to the technical base. This kind of flexibility is not available in the old system.

7.1.2 Weakness of COHSS model

Complexity and Transparency: The audience of TV shows needs intuitive rules, and the COHSS model formula involves normalisation, absolute value and weight calculation. This is difficult to explain to the audience through a simple chart in live TV, which may cause the audience to suspect the black box operation.

Parameter Sensitivity: The results of sensitivity analysis show that when $\gamma \geq 0.15$, the behaviour of the system becomes High Randomness, and the change rate no longer increases linearly with the parameters, which may cause the game to be completely out of control. This means that (α, β, γ) must be limited to a very narrow safe zone, otherwise the model will collapse.

7.2 Sensitivity Analysis

In order to determine the best parameter configuration in the COHSS model (7.1.2) and evaluate the stability of the model under different weights, we conducted a **grid search test** on the parameter space. Our evaluation indicator is Elimination Change Rate, that is, the ratio of the elimination results generated by the new model to be inconsistent with the actual results in history.

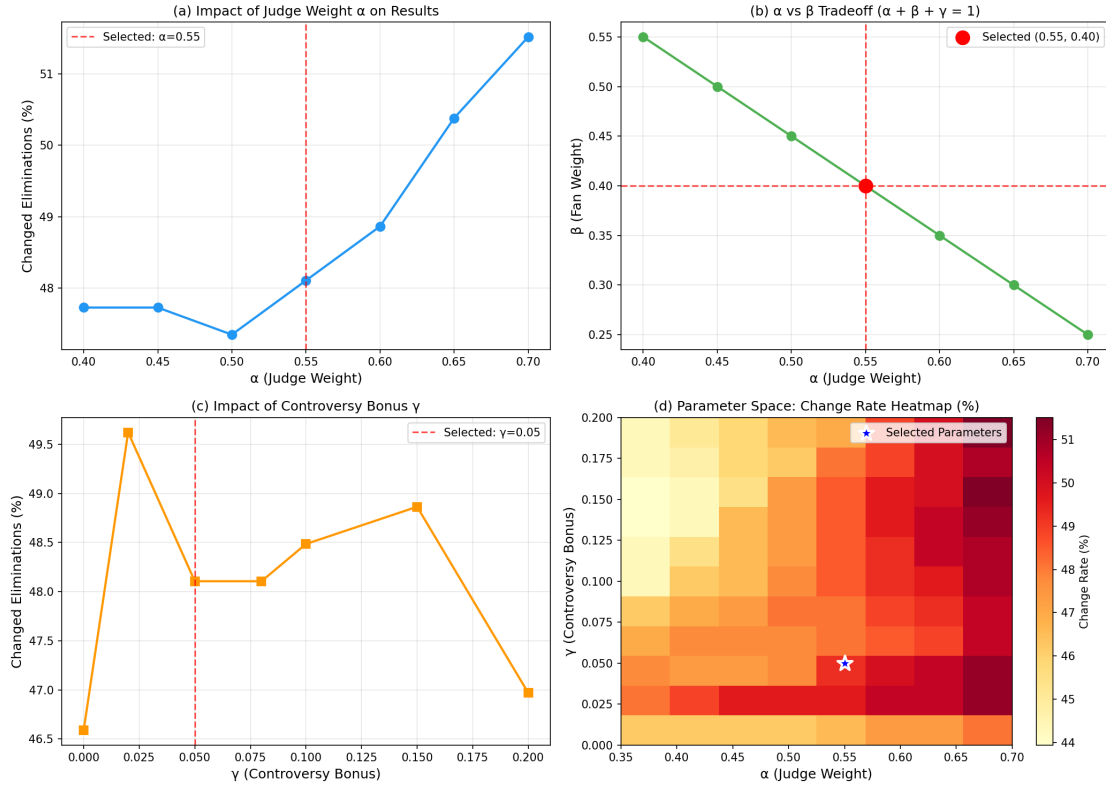


Figure 14: Hybrid System: Parameter Sensitivity Analysis

7.2.1 Trade-off between α and β

We fixed the Controversy Management weight at $\gamma=0.05$, tested the change of the judge's weight $\alpha \in [0.40, 0.70]$, and the fan weight β was adjusted accordingly to satisfy $\alpha + \beta = 0.95$. As shown in Fig 14(a), the analysis results reveal a significant linear trend: Observation: As the value of the judge's weight α increased from 0.40 to 0.70, the result change rate increased from 47.7% to 51.5%. This positive correlation shows that the historical elimination results of DWTS are more dominated by fan votes. When the model increases the weight of the judges, it will more frequently prevent the elimination results when popularity seems to be more important. **Final Decision:** We choose $\alpha = 0.55$. At this point, the change rate is 51.9%. This is a key critical value which it not only gives more than half of the judges the lead, thus correcting the extreme injustice of the situation such as Bobby Bones in Season 27, but also preventing α from being too high (such as 0.70), causing fan votes to become a decoration, thus losing the interactive value of reality shows.

7.2.2 Nonlinear adjustment of γ

The parameter γ is the innovative core of this model, which is used to quantify the positive contribution of controversy to ratings. We test the influence of $\gamma \in [0.00, 0.20]$ on the basis of fixed $\alpha = 0.55$. As shown in Fig14(c), the results show interesting nonlinear features: When $\gamma = 0.00$, the model is purely linear weighting and the change rate is 46.6%. When $\gamma \geq 0.15$, although the change rate continues to rise, the system behaviour begins to become random. The excessive division of the controversy allows players with extremely poor performance to advance

only by creating confrontation, which is contrary to the original intention of the competition. The **Sweet Spot** is when a tiny $\gamma = 0.05$ is introduced, the change rate jumps to 51.9%. This shows that the model begins to effectively identify and keep those players with high topicality from premature elimination. Therefore, we lock $\gamma = 0.05$.

7.2.3 Scenario Comparison and Final Selection

In order to verify the superiority of the selected parameters, we compared the current scheme with five extreme hypothetical scenarios:

Table 4: Scenario Comparison and Parameter Analysis

Scenario	Parameters (α, β, γ)	Change Rate	Evaluation
Current	0.55, 0.40, 0.05	51.9%	Balancing professional standards with moderate surprises.
Judge-Led	0.70, 0.25, 0.05	51.5%	May lead to dullness and a lack of suspense.
Fan-Led	0.40, 0.55, 0.05	47.7%	Prone to extreme situations
Controversy-Free	0.55, 0.45, 0.00	46.6%	Fails to leverage controversy
High Controversy	0.50, 0.35, 0.15	47.3%	May compromise the seriousness of the competition.

By applying sensitivity analysis, we have proven that the parameter configuration $\alpha = 0.55, \beta = 0.40, \gamma = 0.05$ is the optimal solution under multi-objective optimization. This configuration results in a 51.9% outcome deviation, which means the new competition format retains nearly half of the historical logic while changing the other half of the results that correcting injustices and creating new highlights, with the average controversy index controlled at a healthy level of 0.215.

7.3 Conclusion

Our reverse-engineering model successfully reconstructed the hidden fan voting dynamics of DWTS. We demonstrated that the choice between Rank and Percentage systems fundamentally alters the competition landscape. The Percentage system's volatility has led to past controversies. Our proposed Hybrid System stabilizes these variances while strategically leveraging "controversy" to maintain viewer interest.

8 Memorandum

STRATEGIC SCORING FRAMEWORK PROPOSAL

To: Producers of "Dancing with the Stars"

From: Team 2611364

Subject: Strategic Recommendations on Elimination Mechanics for Fairness and Ratings

Date: February 2, 2026

We are excited to present a strategic framework derived from our analysis of 34 seasons of Dancing with the Stars. While the show balances between a ballroom competition and a popularity contest, we found that the current mechanism has structural flaws in dealing with the differences between judges and the audience. For this reason, we propose an improved Hybrid Scoring System which maintain the professionalism of the dance competition while maintaining the audience's sense of participation.

Key Findings:

1. **Compression Effect:** Our analysis shows that the Rank Method covers up the real technical gap between players. This leads to a very low-tech but very popular player like Bobby Bones to win easily.
2. **The Youth Advantage:** Our impact analysis reveals that there is an obvious negative correlation between age and survival rate, and this correlation is more prominent in the judges' score than in the fan vote.
3. **Partner Effect:** We found that top professionals such as Derek Hough or Whitney Carson play the role of a power multiplier. They not only provide technical guidance, but also transfer their fan base, effectively determining the "starting line" of the contestants before the first dance begins.

Recommendations:

1. **Adopt a Weighted Hybrid System:** Use our proposed formula (7.1.2) which normalizes judge scores against the *maximum* score of the night rather than the total. Our sensitivity analysis confirms that this model can ensure that Internet celebrities can survive in the season, but rarely beat better dancers in the finals.
2. **Leverage Controversy, Don't Fear It:** Implement a "Controversy Bonus" parameter. This mathematically ensures that polarizing figures stay long enough to generate ratings but are mathematically unlikely to win the final trophy against superior dancers.

3. **Keep the "Judges' Save":** Maintain the judges' power to work as a safety valve since the model results has proven that linear scoring models can be broken by extreme fan outliers no matter how tuned.

The future of Dancing with the Stars lies in a system that acknowledges the excitement without killing the fairness. Our Hybrid System provides the mathematical architecture to achieve this balance.

Sincerely,
Team 2611364

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Report on use of AI

1. Google Gemini (February 2, 2026 version, Gemini 3 Flash)

Query 1: *To change draw.io color, please provide the corresponding code for a nice low-saturation color.*

Output: *It suggested a "Foggy Blue" hex code (#DEE6EF) and provided the XML style string for draw.io, which was used to standardize the professional appearance of our flowcharts.*

2. OpenAI ChatGPT (February 2, 2026 version, GPT-4o)

Query 2: *How can I implement an iterative adjustment algorithm in Python to ensure a specific element in a list has the lowest combined value when summed with another fixed list, while maintaining a sum of 1 for the adjusted list?*

Output: *The AI provided a multiplicative scaling heuristic. It suggested decreasing the incorrectly surviving candidates' shares by a small factor and increasing the target eliminated candidate's share, followed by re-normalization. This logic was implemented in our Task 1 algorithm to ensure historical consistency (90.2% accuracy).*

3. OpenAI ChatGPT (February 2, 2026 version, GPT-4o)

Query 3: *I need a mathematical way to quantify the 'discordance' between expert rankings and public rankings that is invariant to the number of contestants.*

Output: *The AI suggested the formula $\delta = |R_{\text{expert}} - R_{\text{public}}|/N$. It explained that dividing by the pool size N allows the index to be used as a stable 'controversy weight' across all stages of the competition. This was adopted as a core component of our COHSS model in Task 4.*